CRESS: Cross-Category Reliability Estimation with Source Support for Low-Storage Few-Shot Industrial Anomaly Detection

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ABSTRACT

Few-shot industrial anomaly detectors using frozen foundation features can accurately rank defects, but their thresholds may not maintain consistent false-alarm rates across categories or input shifts. To address this, we audit reliability separately from ranking. Our framework combines a low-storage DINOv2 PCA residual with label-free leave-one-image-out (LOIO) p-values from k normal supports. Evaluating 15 MVTec and 12 VisA categories under four corruption types, our audit reveals a resolution floor: target-only alarms cannot trigger below $\alpha = 1/(k+1)$. Empirical normal p-values range from conservative on VisA at $k = 4$ to anti-conservative on corrupted MVTec. We assess Cross-category Reliability Estimation with Source Support (CRESS), a source-assisted thresholding and certification method using disjoint reference, proposal, and certification categories. Across a frozen grid of four k values, five seeds, five conditions, within-dataset tests, and MVTec-to-VisA/MPDD transfer, category-level Hoeffding certification consistently selects the zero threshold in all 960 gate cells. This outcome is not solely due to Hoeffding looseness: with all observed category losses zero, any deterministic, uniformly valid, distribution-free 95% upper bound needs 14 categories at $\alpha = 0.20$. The most favorable frozen allocation requires 22, 46, or 94 categories at $\alpha = 0.20$, 0.10, or 0.05, respectively, while Hoeffding requires 60, 240, or 958. However, only three or four certification categories are available. Fixed-archive image certificates select positive thresholds in 37-60% of settings, but target a different estimand. Historical non-independent CRESS results remain empirical diagnostics rather than certified evidence. Our main contribution is an auditable reliability framework and a quantitative limit on few-category source certification, rather than unconditional target control.

1. Introduction

Industrial anomaly detection (AD) is typically deployed under an asymmetric data regime: normal examples are available, while abnormal examples are rare, diverse, and costly to annotate. This makes few-shot AD practically important. In this setting, an inspection system receives only k normal support images for a target category, often with $k \in \{1, 2, 4, 8\}$, and must detect both image-level and pixel-level defects at test time.

Recent methods have made this regime much more effective by using frozen visual foundation models. DINOv2 [1] patch features can be stored in a normal memory bank and queried by nearest-neighbor distances, as in PatchCore-style methods and AnomalyDINO [2, 3]. Alternatively, frozen patch features can be compressed into a normal PCA/subspace model, as in SubspaceAD [4]. These approaches provide strong anomaly ranking without training a large task-specific backbone.

However, ranking is not the same as reliable decision support. A raw nearest-neighbor distance or PCA residual can produce high AUROC while still being poorly calibrated: a score threshold may work on one class, but the same numerical score may not carry the same false-alarm meaning under another class, dataset, or corruption. This matters in industrial deployment because operators need to know not only which samples are suspicious, but also whether an alarm probability is trustworthy. Prior work has already shown that DINOv2-based few-shot AD can be poorly calibrated and adversarially fragile [5]. Reliability is therefore not a cosmetic post-processing issue; it is part of the deployment problem.

We address this gap by separating anomaly ranking from reliability estimation. The resulting questions are about the learning system itself, not about any one factory: what false-alarm behavior can a decision rule calibrated on k examples of a frozen representation guarantee, at what resolution, and how does that guarantee fail when the representation's inputs shift? The ranking path remains a frozen DINOv2 PCA/subspace detector: patch residuals are aggregated into a raw anomaly score used for AUROC, AP, and localization. A separate reliability path maps this evidence into calibrated or probability-like outputs. The main target-only route uses leave-one-image-out (LOIO) conformal calibration over the few normal support images. Each support image is scored as if held out, producing normal nonconformity values; a test image then receives a conformal p-value and a reliability score $1 - p$. This view is not a supervised anomaly posterior, but it is statistically interpretable and derived without target anomaly labels.

Each ingredient we build on—DINOv2 features, PCA residuals, Platt-family calibration, and conformal anomaly detection—has an established literature, and our contribution begins after them. What that literature leaves open is a structural question about the few-shot regime itself: a conformal alarm calibrated on k support images cannot fire below $\alpha = 1/(k+1)$ at all, corruption shift can make its observed

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false-alarm rate anti-conservative, and a statistically valid source-category certificate can itself be vacuous when only a few categories are available. The central contribution combines an exact resolution-floor diagnosis, an empirical audit under shift, and Cross-category Reliability Estimation with Source Support (CRESS), a source-assisted thresholding and certification procedure evaluated under a strict nested protocol. The strict experiment does not validate the hoped-for new-category claim: all category-certified thresholds fall back to zero. That negative result is predicted both by the declared Hoeffding rule and by a broader all-zero lower bound for deterministic distribution-free category-risk confidence bounds; it is reported alongside, rather than replaced by, fixed-archive image sensitivity and historical empirical results.

The lesson for learning systems is general even though the testbed is industrial: finite calibration limits both the resolution of target-only alarms and the feasibility of source-category certification. The paper makes four contributions, the first two central:

- We make the few-shot resolution limit of conformal alarms explicit through an attainable-alpha analysis: with k supports no alarm can fire below $\alpha = 1/(k+1)$, and empirical false-alarm rates must be read against the attainable grid, not the nominal level. Cluster-aware Monte Carlo diagnostics of normal p-values then localize when and in which direction the exchangeable-null pattern fails under corruption shift—conservative on VisA, anti-conservative on corrupted MVTec.

- We formulate CRESS with disjoint reference, proposal, and certification categories and evaluate it under a strict nested protocol. Besides the Hoeffding-specific feasibility condition, we prove that any deterministic uniformly valid distribution-free upper bound based only on iid category losses has an all-zero floor $1 - \beta^{1/n}$; even without multiplicity, 95% certification at $\alpha = 0.20$ requires 14 categories. The frozen experiment spans four k, five seeds, five conditions, MVTec, VisA, and MVTec-to-VisA/MPDD transfer. Category certification returns the zero threshold in all 960 gate cells because only three or four certification categories are available; image-level fixed-archive sensitivity is reported separately and never relabeled as a new-category result.

- We formulate a decoupled pipeline: frozen DINOv2 PCA residuals provide ranking, while a label-free layer provides Vector/Shift-Aware calibration and LOIO p-values [6] from the normal supports. Across the historical MVTec and VisA corruption studies, LOIO lowers ECE against most tested calibrators without changing the raw ranking; operational metrics, rather than ECE, carry the central interpretation.

- We report accuracy-storage, transfer calibration, pixel-level, robustness, prevalence-stress, and routing diagnostics, including negative findings that prevent overclaiming. The framework is not MVTec AUROC SOTA, nor is it adversarially robust; ECE on conformal-derived scores is strongly prevalence-sensitive, and the historical CRESS gate is not an independent certificate.

The remainder of this paper is organized as follows. Section 2 positions the work against few-shot detectors, calibration, and conformal anomaly detection. Section 3 presents the decoupled pipeline, the LOIO conformal route, the attainable-alpha analysis, and CRESS. Section 4 describes datasets, protocols, and claim rules. Section 5 reports calibration comparisons, uniformity tests, operational false-alarm behavior, CRESS evaluations with their pre-specified gate, and diagnostics. Section 6 discusses limitations, and Section 7 concludes.

2. Related Work

2.1. Few-Shot Industrial Anomaly Detection

Classical industrial AD often assumes normal-only training data and detects deviations at test time. Patch distribution, self-supervised, and reconstruction-discriminative families such as PaDiM, CutPaste, and DRAEM provide important context for this literature [7–9]. PatchCore [2] made memory-bank patch retrieval a strong baseline by storing representative normal features and scoring test patches by nearest-neighbor distance. AnomalyDINO [3] updates this paradigm for few-shot inspection with frozen DINOv2 representations. The few-shot AD literature has also produced registration-based adaptation (RegAD [10]), graph-structured memory (GraphCore [11]), fast feature reconstruction (FastRecon [12]), vision–language zero-/few-shot scoring (WinCLIP [13]; CLIP–SAM collaboration [14] and cross-modal prompt regularization [15] continue this line), and in-context residual learning (InCTRL [16]). These methods are strong rankers, but the reference state can remain nontrivial at inference time and the raw score scale is not a calibrated probability; calibration under shift is also known to degrade for deep models generally [17]. Our work is complementary: it does not compete on ranking but asks how any such few-shot detector should expose decision reliability.

2.2. Frozen-Feature Subspace Detectors

Subspace approaches replace explicit patch retrieval with a compact normal subspace. SubspaceAD [4] shows that frozen DINOv2 patch features and PCA/subspace residuals are already strong for few-shot anomaly ranking; the ranking mechanism is therefore settled prior art. Our contribution starts after the residual is computed: how should a low-storage subspace detector expose reliability under low-shot and shift, and what can its alarms guarantee?

2.3. Calibration and Robustness in Few-Shot AD

Calibration asks whether predicted confidence agrees with empirical correctness [18, 19]. Standard post-hoc calibrators include Platt scaling [18], temperature scaling [19], histogram binning [20], and isotonic regression [21]; we compare against all of them under a shared label-free protocol. In anomaly detection this is subtle because labels are sparse, anomaly types are open-ended, and the score distribution changes across classes. Reliability analyses of DINOv2-based few-shot AD report poor ECE and strong FGSM fragility [5]. We therefore evaluate ECE, Brier score, and NLL separately from AUROC/AP, and report adversarial results, including FGSM-style perturbations [22], as diagnostics rather than robustness improvements.

2.4. Conformal Prediction and Reliability Routing

Conformal prediction converts nonconformity scores into p-values under exchangeability assumptions [23–25]. This is attractive for normal-only few-shot AD because support normal images can define a reference distribution without requiring real anomalies. Conformal anomaly detection itself is established: leave-one-out, bootstrap, and cross-conformal anomaly detectors have been analyzed directly [6], weighted conformal detection in low-data regimes with resolution and effective-sample-size analyses exists [26], general nonconformity toolkits are available [27], and few-shot conformal prediction with auxiliary tasks addresses the small-calibration-set problem [28]. Conformal AD, LOIO calibration, and weighted conformal p-values are thus established mechanisms, and our contribution begins where they stop in the few-shot regime: at the resolution floor $1/(k+1)$ that no target-only k-shot conformal alarm can cross and at the independent-category requirement of a simultaneous source certificate. We prove the resolution barrier and both Hoeffding-specific and distribution-free all-zero category-feasibility bounds, audit corruption-shift deviations empirically, and evaluate CRESS thresholding below the target-only floor without target anomaly labels. Gated-expert models, including SAGE-style shared/specialized routing [29], motivate the broader idea of choosing conservative or shift-specialized reliability paths. Our framework uses this routing lens, but its main target-only route is intentionally simple: fixed LOIO conformal reliability. Learned or weighted gates are treated as ablations unless they improve reliability without test-label expert selection.

3. Method

3.1. Problem Setup

For each category c, we observe a support set $\mathcal{S}_c = \{x_i\}_{i=1}^k$ containing only normal images. The test set contains normal and anomalous images, but anomaly labels are used only for evaluation, not for fitting the main detector or calibrator. The goal is to produce: (i) a raw anomaly ranking score $s(x)$, (ii) a probability-like reliability score $\hat{p}(y = 1 \mid x)$ or conformal anomaly confidence, and (iii) optional pixel-level maps. Figure 1 gives an overview: a frozen backbone feeds a ranking lane that is never modified, a label-free target-only reliability lane whose conformal alarms hit the attainable-alpha floor, and a CRESS lane that attempts source-assisted operation below that floor but returns a zero threshold when certification is insufficient.

3.2. Frozen Patch Features and Subspace Ranking

We extract patch features with a frozen DINOv2 ViT-S/14 backbone. Let $z_{ij} \in \mathbb{R}^d$ denote patch j from image i. A PCA/subspace model is fit on support patches. With support mean μ and retained principal directions $U \in \mathbb{R}^{d\times m}$, the patch residual is

$$r(z) = \left\|(z-\mu) - UU^\top(z-\mu)\right\|_2 . \tag{1}$$

Patch residuals form an anomaly map. The image-level ranking score is a high-residual aggregation,

$$s(x) = \mathrm{Agg}_j \; r(z_j), \tag{2}$$

where Agg is the patch maximum in the clean accuracy–storage benchmark and the mean of the top-1% patch residuals (ρ=0.01) in the conformal pipeline; the top-fraction variant is a smoothed maximum that stabilizes the LOIO calibration scores. We state this explicitly because the two protocols use different aggregators, and Section 5 reports an aggregation-sensitivity ablation. Importantly, this raw score is the quantity used for AUROC and AP. Calibration routes do not change the ranking score.

3.3. Vector and Shift-Aware Calibration

The first reliability family is post-hoc logistic calibration. A scalar Platt scaler maps $s(x)$ to a probability, while Vector Platt uses a richer descriptor

$$\phi(x) = \left[s_{\mathrm{pca}}(x),\; s_{\mathrm{head}}(x),\; |\tilde{s}_{\mathrm{pca}}(x) - \tilde{s}_{\mathrm{head}}(x)|\right], \tag{3}$$

where s_{head} is a lightweight synthetic-anomaly head score and tildes denote normalized scores. The head is a small MLP (384→256→1, dropout 0.1) trained with binary cross-entropy to separate support normal patches from synthetically perturbed patches; a patch perturbation resamples a support patch and adds Gaussian noise plus a sign shift in feature space. Its image-level score is the maximum patch probability. The head participates only in the calibration descriptor $\phi(x)$, never in the anomaly ranking. The calibrated probability is

$$\hat{p}_\theta(x) = \sigma(w^\top \phi(x) + b). \tag{4}$$

Shift-Aware Platt augments $\phi(x)$ with five shift descriptors: the maximum and mean distance of test patches to the support feature center (scaled by the support 95th-percentile norm), the mean and standard deviation of patch residual scores, and a residual concentration ratio (max over mean). Weighted Platt reweights calibration samples by density-ratio estimates from a logistic domain classifier, following

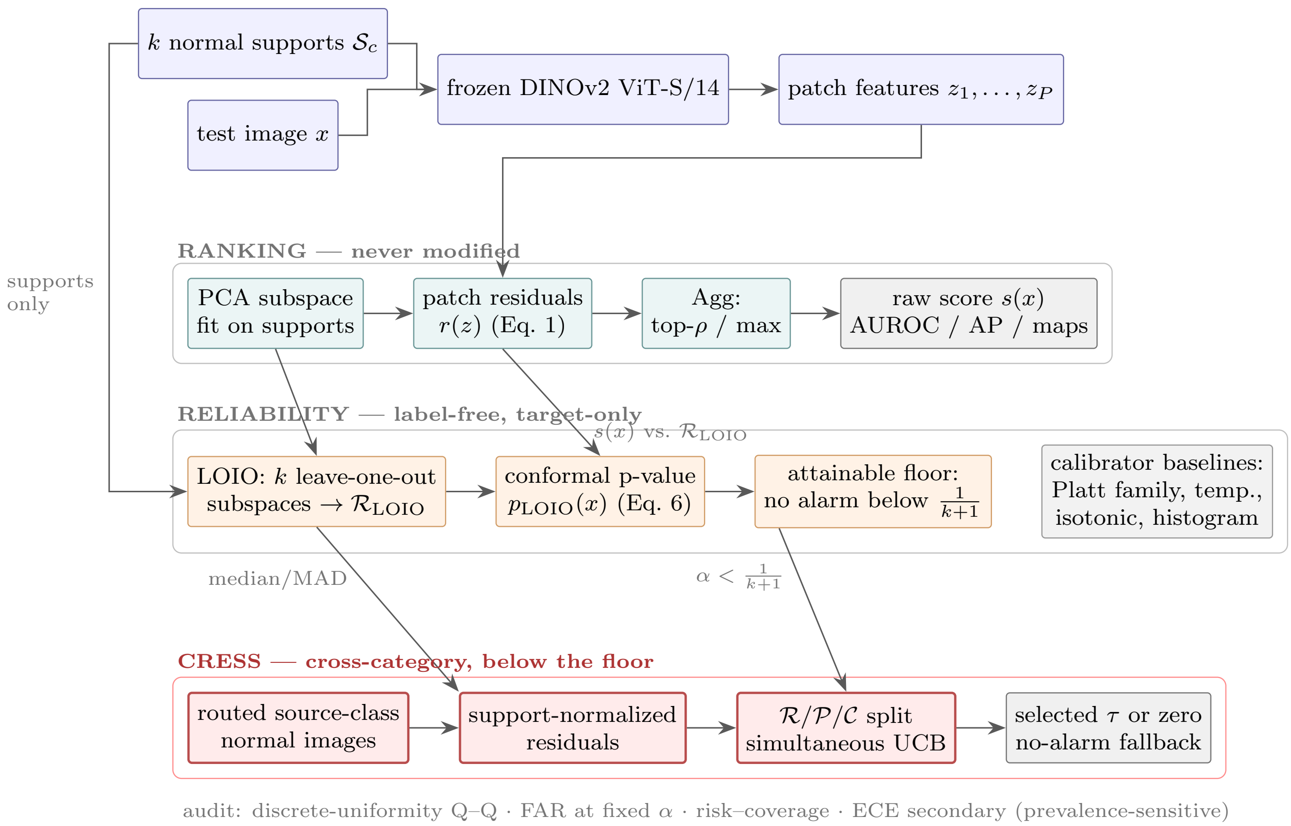

Figure 1: Reliability-framework overview. A frozen DINOv2 backbone feeds two decoupled paths: the *ranking* lane (PCA residuals, never modified; used for AUROC/AP and localization) and the label-free target-only *reliability* lane (LOIO conformal p-values from the k supports, with Platt-family and standard scalar calibrators as baselines). Target-only alarms are structurally silent below $1/(k+1)$. The strict CRESS lane pools routed source-class normals, separates reference, proposal, and certification classes, and selects a threshold only when its simultaneous source-domain upper bound passes; otherwise it returns a no-alarm threshold.

weighted conformal prediction under covariate shift [30]. These variants are useful for structured shifts but can over-adapt, motivating a conservative conformal route.

Calibrator training protocol and label source. All calibrators in the main protocol are trained without any real anomaly labels (mode `normal_synthetic`). The calibration set consists of the k support normal images (label 0) and an equal number of synthetic images (label 1) obtained by replacing a random 25% of a support image's patch features with perturbed support patches as above. Vector-family calibrators fit a multivariate logistic model on $\phi(x)$ by gradient descent with L_2 regularization; the memory-bank and plain-subspace baselines, which expose no calibration descriptor, use scalar Platt on the raw score under the same synthetic protocol. The standard scalar calibrators evaluated in Section 5—temperature scaling [19], isotonic regression [21], and histogram binning [20]—are fit per class on exactly the same calibration set, mapping the raw score $s(x)$ to a probability; since raw residuals are not logits, temperature scaling is applied after centering at the midpoint of the two calibration class means, $\hat{p}(x) = \sigma((s(x)-c)/T)$, with only T learned. An `anomaly_val` mode that calibrates on 10% held-out real anomalies is reported only as an explicitly labeled upper bound. This symmetric, label-free protocol is what makes the LOIO-versus-calibrator comparison fair: every family sees exactly the same k support normals and no test information.

3.4. LOIO Conformal Reliability

For $k \geq 2$, the target-only route constructs normal calibration residuals by leave-one-image-out support splitting. For each support image x_i, a subspace is fit on $\mathcal{S}_c \setminus \{x_i\}$ and the held-out image receives a residual score $s_{-i}(x_i)$. The resulting set

$$\mathcal{R}_{\text{LOIO}} = \{s_{-i}(x_i) : x_i \in \mathcal{S}_c\} \tag{5}$$

serves as normal nonconformity evidence. For a test image x, the conformal p-value is

$$p_{\text{LOIO}}(x) = \frac{1 + \sum_{r \in \mathcal{R}_{\text{LOIO}}} \mathbf{1}\{r \geq s(x)\}}{1 + |\mathcal{R}_{\text{LOIO}}|}. \tag{6}$$

We report $p_{\text{conf}}(x) = 1 - p_{\text{LOIO}}(x)$ as a probability-like anomaly reliability view. It should be interpreted as relative extremeness against held-out support normals, not as a fully supervised posterior probability. Weighted conformal replaces the uniform count by density-ratio weights [30] and reports effective sample size; empirically, it is less stable than LOIO on the main benchmark.

Exchangeability caveat. The construction above is deliberately asymmetric: each calibration residual $s_{-i}(x_i)$ comes from a subspace fit on $k-1$ images, while the test score $s(x)$ uses all k. The two statistics are therefore not exactly exchangeable, and Eq. (6) is a leave-one-out approximation in the spirit of cross-conformal and jackknife+ inference [6, 23, 31] rather than a split-conformal p-value with a formal finite-sample guarantee [32, 33]. We handle this empirically on two fronts. First, a matched-LOIO audit scores held-out support and test images under the same fold-specific subspaces (pairing $s_{-i}(x)$ with r_{-i}, the jackknife+ pairing): it does not improve the operational false-alarm/power trade-off (Section 5), so the simpler construction is retained. Second, rather than assuming validity, we audit it directly: Section 5 reports discrete-grid diagnostics of the normal p-values. Their reference distribution is simulated under an idealized exchangeable null while preserving shared-calibration dependence within class–seed clusters; they localize empirical deviations but are not a proof of exchangeability.

3.5. Attainable Alpha and the Few-Shot Resolution Floor

Proposition 1 (attainable-alpha floor). For any finite calibration scores $r_1, \dots, r_k$, a p-value of the form in Eq. (6) belongs to $\{j/(k+1) : j = 1, \dots, k+1\}$. Hence the alarm $\mathbf{1}\{p(x) \le \alpha\}$ is identically zero for $\alpha < 1/(k+1)$.

Proof. The exceedance count $\sum_i \mathbf{1}\{r_i \ge s(x)\}$ is an integer between 0 and k. Adding one and dividing by $k+1$ gives the stated grid and lower bound. No exchangeability assumption is needed for this algebraic claim. □

At k=4 this floor is 0.2; at k=8 it is $1/9 \approx 0.111$. We make the resolution limit explicit with an attainable-alpha analysis that reports, for each nominal α, the nearest attainable grid point and the empirical false-alarm rate achieved there. This exact structural fact defines the regime in which cross-category support may be useful; it is not presented as the algorithmic novelty by itself. The textbook remedy for discreteness is a smoothed (randomized) conformal p-value, which uses an independent uniform tie-break and is uniform under the corresponding exchangeability assumptions. Section 5 evaluates it as a baseline: randomization crosses the floor but does not by itself address the observed support–test shift.

3.6. Cross-Category Reliability Estimation with Source Support (CRESS)

The attainable-alpha floor motivates the proposed source-assisted procedure. In multi-category factories, normal images from *other* categories (source classes) are plentiful even when the target category has only k normal supports. CRESS uses them without target anomaly labels. First, target and source residuals are made comparable by support normalization: each raw score is centered and scaled by the robust median/MAD statistics of its own class's LOIO support residuals. This normalization is an empirical comparability device, not a proof that class distributions coincide. In matched-condition mode, source images share the target's corruption condition and condition metadata is therefore required. Clean-source always uses clean normals. The condition-agnostic ablation takes the median score across all available condition views of each base source image, so repeated corruptions are not counted as independent observations; deterministic mismatched-condition routing is only a negative control. Pooling source normals gives finer p-value resolution than the k target supports and permits thresholds below $1/(k+1)$.

3.7. Nested Source Certification and Transfer Boundary

For a fixed target, source classes are partitioned before certification into reference classes $\mathcal{R}$, proposal classes $\mathcal{P}$, and certification classes $\mathcal{C}$. Scores from $\mathcal{R}$ define conformal p-values. The distinct p-values on $\mathcal{P}$ propose at most M positive candidate thresholds. For candidate τ, certification unit i has loss $L_i(\tau) \in [0, 1]$. An image unit uses its Bernoulli alarm indicator and targets risk for the fixed source-image mixture. A class unit averages alarms within a held-out source class and targets a new draw from a source-category meta-population. Let A be the number of reported α levels. For the class unit with n classes, CRESS uses

$$U_{\mathrm{H}}(\tau) = \min\left\{1,\ \bar{L}(\tau) + \sqrt{\frac{\log(2AM/\delta)}{2n}}\right\}. \tag{7}$$

For independent Bernoulli image alarms, we use the exact one-sided binomial (Clopper–Pearson) upper bound with tail probability $\delta/(2AM)$. The factor $2A$ allocates the family error across both units and all reported levels; M allocates it across candidates. CRESS selects the largest proposed τ whose corresponding upper bound is at most α. If none passes, $\tau = 0$ gives a deterministic no-alarm fallback. Reference, proposal, and certification classes are disjoint; target test images and labels enter none of these stages.

Proposition 2 (conditional source certificate). Conditional on the reference and proposal stages, suppose certification units are independent draws from the declared source-unit population: Bernoulli losses for the image analysis, or bounded losses in $[0, 1]$ for the class analysis. Then, with probability at least $1 - \delta$, simultaneously across the A levels and both unit definitions, the corresponding source risk of every selected threshold is at most its α.

Proof. For a fixed level, unit, and candidate, the one-sided Clopper–Pearson construction or Hoeffding's inequality fails with probability at most $\delta/(2AM)$. A union bound over M candidates, A levels, and two units gives total failure probability at most δ; no independence between these bounds is required. Selection occurs only on this simultaneous event, so each passing candidate has risk at most its level. The zero fallback has zero alarm risk. □

Proposition 3 (Hoeffding feasibility). Under the category-level Hoeffding rule above, a positive candidate can pass at level α only if

$$n \geq \frac{\log(2AM/\delta)}{2\alpha^2}. \tag{8}$$

Proof. Because $\bar{L}(\tau) \geq 0$, the unclipped upper bound is at least $\sqrt{\log(2AM/\delta)/(2n)}$. Requiring it to be no larger than α and rearranging gives Eq. (8). Any positive empirical loss only increases the requirement. □

Proposition 4 (distribution-free all-zero lower bound). Let $\beta \in (0,1)$ and let $L_1, \dots, L_n$ be iid losses from an arbitrary distribution P on $[0,1]$, with mean μ_P. Let $U_n : [0,1]^n \to [0,1]$ be any deterministic upper confidence bound satisfying

$$\inf_P \Pr_{P^n}\{\mu_P \leq U_n(L_1, \dots, L_n)\} \geq 1 - \beta.$$

Then

$$U_n(0, \dots, 0) \geq 1 - \beta^{1/n}. \tag{9}$$

Consequently, even after observing zero loss in every certification category, $U_n \leq \alpha$ is possible only if

$$n \geq \frac{\log \beta}{\log(1-\alpha)}.$$

Proof. Write $u = U_n(0, \dots, 0)$ and suppose, for contradiction, that $u < 1 - \beta^{1/n}$. Choose a Bernoulli distribution with mean q such that $u < q < 1 - \beta^{1/n}$; Bernoulli laws are a subclass of distributions on $[0,1]$. Its all-zero sample has probability $(1-q)^n > \beta$. On that event $U_n = u < q = \mu_P$, so the bound fails with probability greater than β, contradicting uniform $(1-\beta)$ coverage. Rearranging $1 - \beta^{1/n} \leq \alpha$ gives the sample-size condition. □

Proposition 4 shows that the frozen failure is not solely an artifact of Hoeffding's looseness. Even for one fixed candidate, one level, no image-unit co-reporting, and no multiplicity penalty ($\beta = \delta = 0.05$), any deterministic uniformly valid distribution-free bound based only on iid category losses needs at least 14, 29, and 59 categories at $\alpha = 0.20$, 0.10, and 0.05. Under the frozen equal allocation $\beta = \delta/(2AM)$ with $A = 3$ and the most favorable $M = 1$, the corresponding lower limits are 22, 46, and 94; the declared Hoeffding rule requires 60, 240, and 958. Every requirement exceeds the available three or four certification categories. Table 1 is a design calculation, not an empirical result. Proposition 4 is deliberately scoped: it concerns deterministic, uniformly valid, distribution-free upper bounds that use only iid bounded category losses. Parametric or hierarchical assumptions, informative side data, or randomized confidence procedures fall outside its statement and would require their own validity argument. We do not present this elementary minimax argument as new confidence-bound theory; its role is to rule out a misleading "Hoeffding is merely loose" interpretation of the experiment. The calculation also clarifies why adding source images or repeated support seeds cannot repair a shortage of independent categories: those additions may tighten the fixed-archive image analysis but do not increase category-level n. Conversely, changing the certification unit after seeing the result would change the estimand and is not a valid rescue of the frozen protocol.

Table 1
Best-case minimum independent certification categories at $\delta = 0.05$, assuming zero observed category loss. "DF floor" is the distribution-free lower bound in Proposition 4: first without multiplicity ($\beta = \delta$), then under the frozen allocation $\beta = \delta/(2AM)$ with $A = 3$. "Hoeffding" is Proposition 3.

rule / candidates	$\alpha = .20$	$\alpha = .10$	$\alpha = .05$
DF floor, no multiplicity	14	29	59
DF floor, $M = 1$	22	46	94
DF floor, $M = 5$	29	61	125
DF floor, $M = 20$	35	74	152
Hoeffding, $M = 1$	60	240	958
Hoeffding, $M = 5$	80	320	1,280
Hoeffding, $M = 20$	98	390	1,557

Corollary 1 (conditional target transfer). On the event in Proposition 2, target risk is also at most α if, for every proposed threshold, the target normal alarm probability is no larger than the corresponding source-unit risk. The same follows if the target category is exchangeable with the certification-category meta-population.

The corollary's dominance or category-exchangeability condition is an assumption, not a consequence of robust normalization or matched routing. The two source certificates answer different questions: an image certificate for a fixed archive mixture cannot be relabeled a new-category guarantee, while a class certificate with few categories may be vacuous and cannot be replaced by the image result. The frozen nested results in Section 5 exhibit exactly this category-level failure; the older non-independent results remain labeled historical empirical diagnostics. A direct pooled-source conformal baseline uses the entire routed source pool with threshold α and receives no artificial data restriction; unlike CRESS, it performs no proposal/certification step.

The broader routing component is inspired by the shared-versus-specialized principle in gated-expert systems. Candidate reliability views include Vector Platt, Shift-Aware Platt, weighted Platt, LOIO conformal, and weighted conformal. We evaluate fixed routes, validation-ECE selection, and simple no-label rules based on shift descriptors and effective sample size. Because learned gates did not dominate fixed LOIO across all settings, the main paper reports fixed LOIO as the target-only reference and treats learned gates as diagnostic ablations; it is not called a safe route under distribution shift.

3.8. Selective Diagnostics

Beyond producing a calibrated alarm score, the reliability framework records entropy, expert disagreement, conformal confidence, and effective sample size as selective-risk signals. These quantities are not used to change the anomaly ranking. Instead, they support deployment diagnostics such as abstention, risk-coverage analysis, and manual review of low-confidence alarms.

4. Experiments

4.1. Datasets and Few-Shot Splits

We evaluate on MVTec AD [34] and VisA [35]; the strict external-transfer audit additionally uses MPDD. MVTec AD contains industrial object and texture categories with pixel-level defect masks. VisA contains a broader set of industrial categories and is used as the main full-scale corruption and calibration-shift benchmark. MPDD supplies six metal-part categories for an external MVTec-to-MPDD stress test; within-MPDD category certification is not claimed because its category count is too small for a three-way source split. For each class, support sets are sampled from normal training images with fixed seeds. Unless otherwise specified, $k \in \{1, 2, 4, 8\}$ is used for clean accuracy-storage and strict nested studies, while the earlier conformal corruption tables use $k \in \{4, 8\}$.

4.2. Evaluation Protocols

We organize experiments into six protocol groups. The first five cover clean accuracy/storage, PCA64/PCA128 sensitivity, corruption reliability, representative routing, and attainable operating levels. The sixth is the frozen strict nested protocol: $k \in \{1, 2, 4, 8\}$, seeds 0–4, $\alpha \in \{0.05, 0.10, 0.20\}$, clean plus Gaussian noise, blur, brightness/contrast, and JPEG, at most 120 evaluation images per cell, $\rho = 0.01$, PCA64, $\delta = 0.05$, and at most 20 candidate thresholds. It evaluates MVTec and VisA within-dataset, MVTec-to-VisA, and MVTec-to-MPDD, with matched, clean-source, condition-agnostic, and mismatched negative-control routing. The pre-specified operational gate requires a nonzero category-certified threshold in at least 80% of target cells, target FAR at most $\alpha + 0.02$, nonzero power, at least 80% no-harm, and a positive simultaneous power-gain lower bound. Zero thresholds and failed cells are retained.

4.3. Naming Convention

CRESS denotes only the proposed cross-category source-assisted procedure, not the frozen ranker or the target-only reliability routes. Detector instantiations are identified by their retained subspace dimension, PCA64 (default) or PCA128. Reliability routes are scalar/Vector/Shift-Aware/weighted Platt, temperature scaling, isotonic regression, histogram binning, LOIO conformal, and weighted conformal. The strict nested certification protocol evaluates CRESS; it is not part of the method name.

4.4. Baselines and Ablations

We compare against a controlled nearest-neighbor memory bank using DINOv2, a SubspaceAD-style PCA residual baseline, scalar Platt, Vector Platt, Shift-Aware Platt, weighted Platt, temperature scaling, isotonic regression, histogram binning, LOIO conformal, and weighted conformal. The controlled memory-bank implementation is not an official reproduction of PatchCore or AnomalyDINO; it holds the frozen DINOv2 feature pipeline and data split fixed to isolate the scoring and storage trade-off. In addition, we run the official AnomalyDINO code (ViT-S/14 at 448px, agnostic preprocessing) on both MVTec and VisA (official 1-cls split) for $k \in \{1, 4, 8\}$ with three seeds and report it as a separate accuracy reference. As the vision–language representative we include WinCLIP [13], but only as reported numbers: no official implementation is released, and when we audited the leading community reimplementation under its own protocol (ViT-B-16+240, LAION-400M; zero-shot and 1-shot with its fixed support lists, three runs), it reproduced 71.7/77.4 mean image AUROC on MVTec and 66.0/70.0 on VisA against the reported 91.8/93.1 and 78.1/83.8—a roughly 20-point gap that the reimplementation's own authors also report and that we cannot attribute. This is precisely why our claim rules separate audited rows from reported rows, and WinCLIP therefore enters Table 2 as a cited reference, not an audited baseline. Official SubspaceAD representative runs are included as a novelty guardrail. Their strong performance confirms that frozen DINOv2 subspace residuals are an existing competitive ranking mechanism; the residual is inherited, and this paper's contribution begins at the reliability of the alarms built on it.

4.5. Metrics

We separate ranking metrics from reliability metrics. AUROC and AP are computed from the raw PCA/subspace residual score $s(x)$. Pixel-level localization is measured by pixel AUROC and AU-PRO when masks are available. Reliability is measured operationally by empirical false-alarm rate at fixed α, detection power, alarm precision, discrete uniformity of normal p-values, and risk–coverage curves; expected calibration error (ECE), Brier score, and negative log-likelihood (NLL) are reported as secondary metrics using the selected reliability probability or probability-like conformal view. Storage is measured as retained reference/model state per class. Robustness experiments include corruption shift and a label-aware PCA-residual FGSM surrogate; the latter is used only as a fragility diagnostic.

4.6. Reproducibility and Claim Rules

Experiments were run on a single NVIDIA RTX 5090 GPU with PyTorch 2.12.1 and DINOv2 patch features cached in fp32. For reproducibility, we record the frozen commit, DINOv2 source and weight hashes, environment, dataset counts, per-image predictions, base-image identities, support manifests, corruption parameters, source partitions, candidate-threshold bounds, and SHA-256 checksums. Automated integrity checks cover 1,500 MVTec, 1,200 VisA, and 600 MPDD view cells and flag duplicate cells, missing support statistics, or support–test overlap. Reliability routes cannot claim ranking improvement unless the raw score changes; image-level and category-level certificates cannot substitute for one another.

Table 2
Clean image-level ranking performance and category-specific storage. Controlled DINOv2 NN and PCA64/PCA128 share our feature, split, metric, and storage-accounting protocol and are averaged over classes and five seeds. The official AnomalyDINO reproduction uses three seeds under its released protocol; WinCLIP values are reported in [13]. Bold marks the best value only among common-protocol rows in each dataset/k block. "–" denotes not evaluated, N/C not directly comparable, and N/R not reported.

Dataset	k	Method	AUROC ↑	AP ↑	ECE ↓	MiB ↓
MVTec	1	Official AnomalyDINO	0.965	0.982	–	–
		WinCLIP (reported)	0.931	N/C	N/R	N/R
		Controlled DINOv2 NN	**0.914**	**0.952**	0.277	2.005
		PCA64 (ours)	0.904	0.951	**0.274**	**0.472**
	4	Official AnomalyDINO	0.976	0.984	–	–
		WinCLIP (reported)	0.952	N/C	N/R	N/R
		Controlled DINOv2 NN	**0.942**	**0.967**	0.693	6.000
		PCA64 (ours)	0.937	**0.967**	**0.214**	**0.472**
	8	Official AnomalyDINO	0.980	0.990	–	–
		Controlled DINOv2 NN	**0.948**	0.970	0.693	6.000
		PCA64 (ours)	0.945	**0.972**	**0.154**	**0.472**
VisA	1	Official AnomalyDINO	0.857	0.866	–	–
		WinCLIP (reported)	0.838	N/C	N/R	N/R
		Controlled DINOv2 NN	0.804	0.809	0.434	2.005
		PCA64 (ours)	0.823	0.834	**0.429**	**0.472**
		PCA128 (ours)	**0.834**	**0.842**	0.430	0.566
	4	Official AnomalyDINO	0.912	0.918	–	–
		WinCLIP (reported)	0.873	N/C	N/R	N/R
		Controlled DINOv2 NN	0.862	0.861	0.543	6.000
		PCA64 (ours)	0.870	0.883	0.284	**0.472**
		PCA128 (ours)	**0.885**	**0.895**	**0.278**	0.566
	8	Official AnomalyDINO	0.926	0.929	–	–
		Controlled DINOv2 NN	0.873	0.870	0.533	6.000
		PCA64 (ours)	0.880	0.891	0.205	**0.472**
		PCA128 (ours)	**0.897**	**0.905**	**0.170**	0.566

5. Results

5.1. Ranking Substrate and Storage

Table 2 presents clean image-level ranking performance and category-specific storage. The official AnomalyDINO reproduction (released code, three seeds) achieves the highest image AUROC at each reported k on both datasets: 0.965, 0.976, and 0.980 on MVTec, and 0.857, 0.912, and 0.926 on VisA. Because these results follow the released method's own evaluation protocol, they are external accuracy references rather than direct evidence of superiority under our controlled protocol. WinCLIP is likewise a cited reference: its paper reports AUROC and AUPR at 1/4 shots, but not an AP implementation verified to match ours. The reported AUPR values are 0.965/0.973 on MVTec and 0.851/0.888 on VisA; we therefore mark its AP entries N/C instead of mixing them into the AP comparison. WinCLIP does not report ECE under our calibration protocol, category-specific storage under our accounting, or an 8-shot result. The absence of an official implementation and the community-reimplementation discrepancy are documented in Section 4.

Compared with the controlled DINOv2 nearest-neighbor ranker, the PCA64 residual ranker yields lower AUROC by 0.010/0.005/0.003 on MVTec and higher AUROC by 0.019/0.008/0.007 on VisA at $k = 1/4/8$, respectively, while requiring 0.472 MiB of category-specific state instead of 2.005–6.000 MiB. Accordingly, PCA64 is treated as a compact supporting ranker rather than a ranking contribution or state-of-the-art result. On VisA, PCA128 increases class-averaged AUROC over PCA64 by 0.011/0.015/0.017 at $k = 1/4/8$, with storage rising to 0.566 MiB; this is a descriptive accuracy–storage ablation rather than a new ranking-method claim. The controlled nearest-neighbor rows are not official PatchCore or AnomalyDINO reproductions. Boldface in Table 2 is therefore restricted to comparisons among controlled-protocol rows.

The controlled nearest-neighbor ECE values diagnose a failed scalar calibration fit rather than a failure in ranking. Under the label-free scalar-Platt protocol, exact support-patch self-matches at low k and the 4096-patch subsampling regime at larger k coincide with near-constant test probabilities. On MVTec, ECE is 0.277 at $k = 1$ and 0.693 at $k = 4, 8$, which correspond to the class-averaged prevalence complements; the Brier score is similarly close to ECE. These degenerate probabilities are not used as a competitive calibration baseline.

Storage and compute accounting. Storage is reported in MiB and includes only category-specific state, excluding the shared frozen DINOv2 ViT-S/14 backbone ($\approx$ 22M parameters, $\approx$ 84 MiB in fp32), which is stored once per deployment for all compared frozen-feature methods. The 0.472 MiB PCA64-based reliability state consists of the

Table 3
Target-only LOIO operating points. The floor is the smallest p-value, $1/(k+1)$; "attainable" is the largest p-value grid point not exceeding the nominal α, or zero if none exists. FAR and detection are averaged over classes, seeds, and corruptions.

dataset	k	floor	nominal α	attainable	FAR	detection
VisA (full)	4	0.200	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.200	0.152	0.599
	8	0.111	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.111	0.111	0.537
MVTec (full)	4	0.200	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.200	0.411	0.887
	8	0.111	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.111	0.304	0.907

PCA mean and 64-component basis ($\approx$ 0.095 MiB), the synthetic-anomaly MLP head, and its feature-normalization statistics. Increasing the basis from 64 to 128 components adds 0.094 MiB. The fitted three-feature Vector Platt scaler requires only tens of bytes, which is below the displayed precision. The controlled memory bank stores up to 4096 fp32 vectors of dimension 384, exactly 6.000 MiB at the limit. For cached features, raw PCA-residual scoring requires approximately 1.3 ms per image, compared with 5–13 ms for controlled nearest-neighbor scoring. These timings exclude shared DINOv2 feature extraction and reflect only category-specific raw-score computation. LOIO also requires a one-time setup of k leave-one-out PCA fits per category, each using up to $(k-1) \times 1369$ support patches.

The clean benchmark uses max pooling, while the conformal pipeline uses the top-1% mean. The representative aggregation ablation in Table 15 (Appendix A) shows that both choices are within 0.01 to 0.02 AUROC of the best tested setting. This suggests that pooling does not drive the ranking comparison, but it does not establish invariance outside the tested grid.

5.2. Target-Only Resolution and False-Alarm Behavior

Calibration error does not address the operator's main concern: "if an alarm fires at $p \leq \alpha$, how often is a normal part flagged?" Table 3 highlights the structural resolution limit. For $k = 4$, no p-value lies below $1/(k+1) = 0.2$, so every nominal level below 0.2 raises zero alarms for both datasets. For $k = 8$, the minimum is $1/9 \approx 0.111$. These zeros reflect resolution artifacts from the finite calibration set, not evidence of effective conservative control.

Table 4 assesses the first attainable operating point, $\alpha = 0.20$. The target-only rule is conservative yet effective on VisA (at $k = 4$, FAR 0.14–0.16, detection power 0.58–0.61, and precision about 0.81), but is anti-conservative on corrupted MVTec at $k = 4$. Therefore, increasing α above the resolution floor alone does not ensure stable false-alarm behavior under distribution shift. A numerical implementation detail is also relevant: an fp32 representation of $1/5$ may be 0.20000000298, so exact comparison with 0.2 can inadvertently exclude all floor-level alarms. All reported thresholding applies a small comparison tolerance.

5.3. Strict CRESS Category Certification

The frozen strict result is negative. Among 960 category-level gate cells (four jobs, four routing modes, four k values, three α levels, and five conditions), none meet the required 80% nonzero-threshold rate. Every category-certified target cell receives $\tau = 0$, resulting in all 960 operational gates failing. The lowest observed category candidate UCB is 0.950 on MVTec, 1.000 on VisA, 0.961 for MVTec-to-VisA, and 0.986 for MVTec-to-MPDD, all above the largest tested $\alpha = 0.20$. This outcome is not a GPU or optimization failure. Proposition 3 demonstrates that with $A = 3$, $\delta = 0.05$, and even $M = 1$, a zero-loss Hoeffding gate needs at least 60 categories at $\alpha = 0.20$. Proposition 4 further indicates that this result is not solely a consequence of Hoeffding looseness: even for one fixed candidate without a multiplicity penalty, any deterministic, uniformly valid, distribution-free 95% upper bound based only on iid category losses needs at least 14 all-zero categories to reach 0.20; the frozen family allocation increases this to 22. Only three or four certification categories are available.

The fixed-archive image analysis provides more informative results but addresses a narrower estimand. Positive image-certified thresholds occur in 36.7% of MVTec, 60.4% of VisA, 41.5% of MVTec-to-VisA, and 37.0% of MVTec-to-MPDD settings across the same grid. These rates remain below the 80% operational criterion except in certain individual cases, and repeated corruption views are not considered independent images. Thus, the image-level result serves as a sensitivity analysis: it does not resolve the category-level failure or establish target-category false-alarm control.

Table 4
Target-only LOIO false-alarm rate (FAR), anomaly-detection rate, and precision at the first reported attainable level, α=0.20. Results are pooled over classes and seeds and reported separately by dataset, corruption, and k. Alarms fire at $p \leq \alpha$ using the comparison tolerance described in the text.

dataset	corruption	k	FAR	detection	precision
VisA (full)	blur	4	0.155	0.610	0.806
		8	0.109	0.546	0.841
	bright/contr.	4	0.155	0.595	0.803
		8	0.110	0.533	0.836
	Gauss. noise	4	0.142	0.585	0.813
		8	0.103	0.522	0.843
	JPEG	4	0.157	0.604	0.803
		8	0.123	0.547	0.825
MVTec (full)	blur	4	0.306	0.887	0.872
		8	0.194	0.908	0.916
	bright/contr.	4	0.313	0.887	0.869
		8	0.188	0.908	0.919
	Gauss. noise	4	0.463	0.879	0.816
		8	0.355	0.898	0.855
	JPEG	4	0.410	0.890	0.836
		8	0.307	0.914	0.875

Table 5
Frozen strict nested CRESS audit over four routing modes, $k \in \{1, 2, 4, 8\}$, three α levels, five seeds, and five conditions. "Min UCB" is the smallest observed candidate upper bound. Nonzero is the fraction of target cells receiving $\tau > 0$. Category and image rows target different estimands.

source → target	certification unit	units/cell	min UCB	nonzero	interpretation
MVTec → MVTec	category	4	0.950	0.000	new-category gate fails
VisA → VisA	category	3	1.000	0.000	new-category gate fails
MVTec → VisA	category	4	0.961	0.000	new-category gate fails
MVTec → MPDD	category	4	0.986	0.000	new-category gate fails
MVTec → MVTec	fixed-archive image	75–191	0.039	0.367	sensitivity only
VisA → VisA	fixed-archive image	150–180	0.042	0.604	sensitivity only
MVTec → VisA	fixed-archive image	72–159	0.048	0.415	sensitivity only
MVTec → MPDD	fixed-archive image	81–190	0.040	0.370	sensitivity only

5.4. Conformal Reliability under Corruption

Table 6 presents the full VisA aggregate using a fixed PCA/subspace ranking score. Across 56,000 corruption-view rows, LOIO conformal attains a pooled ECE of 0.0766, with 0.0391 at $k = 4$ and 0.1140 at $k = 8$. Weighted conformal has substantially higher ECE at $k = 4$ and approaches LOIO at $k = 8$ (0.116 versus 0.114). In the full MVTec study (15 classes, $k \in \{4, 8\}$, seeds 0–2, four corruptions, 37,464 rows), LOIO attains an overall ECE of 0.0684 (0.0600 at $k = 4$ and 0.0769 at $k = 8$) while maintaining an image AUROC of 0.912. LOIO has lower ECE than Vector Platt and Shift-Aware Platt in all eight k–corruption settings; for example, at MVTec $k = 4$ under Gaussian noise, ECE is 0.033 compared with 0.233 and 0.270. This is a consistent descriptive pattern over the tested cells, not a confirmatory significance result. ECE remains a secondary metric and does not provide a false-alarm guarantee.

Table 7 (Appendix A) provides VisA per-corruption values, and Figure 3 compares both datasets. LOIO has the lowest ECE in all four VisA $k = 4$ cells and under Gaussian noise at $k = 8$; weighted conformal is lower for the other three VisA $k = 8$ corruptions. Across the MVTec panels, LOIO is the lowest-ECE route in every displayed cell. This cellwise summary is more precise than describing either route as uniformly best.

5.5. Distributional and Calibration Diagnostics

Conformal p-values satisfy $P(p \leq \alpha) \leq \alpha$ for normal data only when exchangeability conditions hold. This represents a false-alarm rate, not a posterior probability. Since a k-shot LOIO p-value is restricted to the grid $\{j/(k+1)\}$, applying a continuous Kolmogorov–Smirnov test against $U(0, 1)$ is not valid. Figure 4 instead compares the empirical CDF of normal p-values to the ideal discrete-uniform reference at each grid point. Deviations are conservative for VisA at $k = 4$ (worst gap -0.10 at 0.4), but VisA at $k = 8$ and MVTec at both k become anti-conservative under corruption. For example, for MVTec $k = 4$ with Gaussian noise, $\hat{F}(0.2) = 0.46$ compared to the reference 0.20. Because rows share calibration sets, classes, and repeated seeds, we use

Table 6
Full VisA conformal reliability under four corruptions, pooled over classes and seeds. The PCA/subspace ranking score is fixed, so AUROC and AP do not change between reliability routes; N is the number of evaluated corruption-view rows. Bold marks the lower ECE, Brier score, and NLL within each group.

Group	Reliability route	N	AUROC	AP	ECE ↓	Brier ↓	NLL ↓
All	LOIO conformal	56000	0.823	0.846	**0.077**	**0.187**	**0.686**
	Weighted conformal	56000	0.823	0.846	0.166	0.244	0.813
k=4	LOIO conformal	28000	0.820	0.842	**0.039**	**0.187**	**0.760**
	Weighted conformal	28000	0.820	0.842	0.216	0.259	0.925
k=8	LOIO conformal	28000	0.828	0.851	**0.114**	**0.188**	**0.612**
	Weighted conformal	28000	0.828	0.851	0.116	0.229	0.701

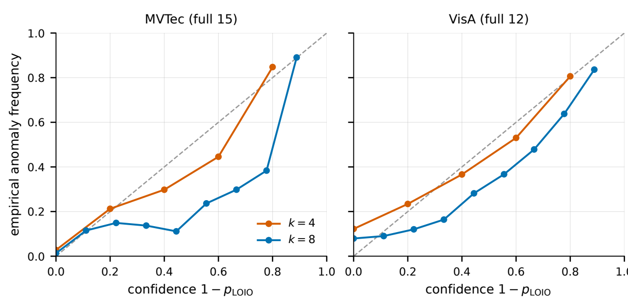

Figure 2: LOIO reliability diagrams on the full MVTec (left) and VisA (right) corruption benchmarks, pooled over classes, seeds, and corruptions. Points occupy the k+1 attainable confidence levels; positions below the diagonal indicate over-confidence.

a cluster-preserving exchangeable Monte Carlo diagnostic rather than an iid confirmatory test. The results suggest that corrupted normal residuals increase relative to clean LOIO calibration scores, but do not identify a causal mechanism. A matched-LOIO audit on representative MVTec $k = 4$ classes increases clean FAR from 0.17 to 0.25 with power remaining essentially unchanged. Therefore, fold asymmetry alone does not explain the observed behavior.

Table 8 presents class, seed, and corruption comparisons for Vector Platt, Shift-Aware Platt, temperature scaling, isotonic regression, histogram binning, and scalar Platt, all under the same label-free protocol. Unlike the pooled ECE in Table 6, these entries first compute ECE within each class–seed–corruption cell and then report the mean and standard deviation across cells; the two summaries therefore need not be numerically equal. LOIO has lower mean per-cell ECE in most comparisons (for example, MVTec $k = 4$: LOIO 0.120 ± 0.054 versus isotonic 0.222 ± 0.088, temperature 0.240 ± 0.104, and histogram binning 0.229 ± 0.099), though the margin decreases at $k = 8$. LOIO has lower ECE than isotonic regression in 55% of MVTec cells and 62% of VisA cells. Against Shift-Aware Platt on VisA $k = 8$, the mean difference is -0.011 and LOIO has lower ECE in 51% of cells. We do not report historical unadjusted significance values because a complete multiplicity-adjusted analysis is unavailable, and we make no confirmatory significance claims. Equal-mass ECE increases LOIO's per-cell error compared with equal-width ECE, highlighting the importance of the operational analyses above.

ECE on $1 - p$ is sensitive to anomaly prevalence, even when ranking remains fixed. On the full VisA dataset, LOIO ECE decreases from 0.404 at 1% anomaly prevalence to 0.149 at 50%, and the ordering between LOIO and weighted methods reverses across this range. As a result, anomaly-balanced ECE cannot guarantee deployment reliability. Throughout this paper, FAR, power, precision, and risk–coverage are used for operational interpretation.

5.6. Historical, Non-Independent CRESS Sensitivity

These analyses were conducted before the strict nested protocol and are included solely as empirical sensitivity checks. Because they reuse source evidence and use pointwise rather than simultaneous intervals, they do not supersede the category-level result in Section 5.3.

Across all 15 label-stratified MVTec classes at $k = 4$ (seeds 0–2, five conditions, 675 evaluation cells), matched-condition CRESS achieves mean power of 0.216 and 0.416 at $\alpha = 0.05$ and 0.10, respectively, with the target-only anchor inactive. Mean FAR is 0.050/0.105/0.216 at $\alpha = 0.05/0.10/0.20$, precision exceeds 0.90, and no-harm rates are 89%/82%/89%. While the historical empirical gate passes overall, some condition-level exceptions persist: JPEG FAR is 0.121 at $\alpha = 0.10$, and Gaussian noise and JPEG reach 0.226/0.235 at $\alpha = 0.20$, exceeding the $\alpha + 0.02$ thresholds.

Supporting tables are provided in Appendix A. Table 9 details the $k = 4$ condition breakdown. At $k = 8$, matched-condition CRESS achieves power of 0.547/0.693 at the two nominal levels below the target-only floor. Its pooled FAR of 0.052/0.121/0.230 and no-harm rates of 80%/74%/81% meet the historical empirical budget only at $\alpha = 0.05$ (Table 10). Smoothed randomized p-values also enable alarms below the deterministic floor but retain the target-only mismatch: on MVTec, FAR is 0.067 under clean data at $\alpha = 0.05$ and increases to 0.115 under Gaussian noise. The historical source route shows similar pooled sub-floor power with closer-to-nominal empirical FAR (Table 11). Within-VisA sources yield FAR of 0.051/0.095/0.192 and sub-floor power of 0.144/0.351. Using MVTec sources instead

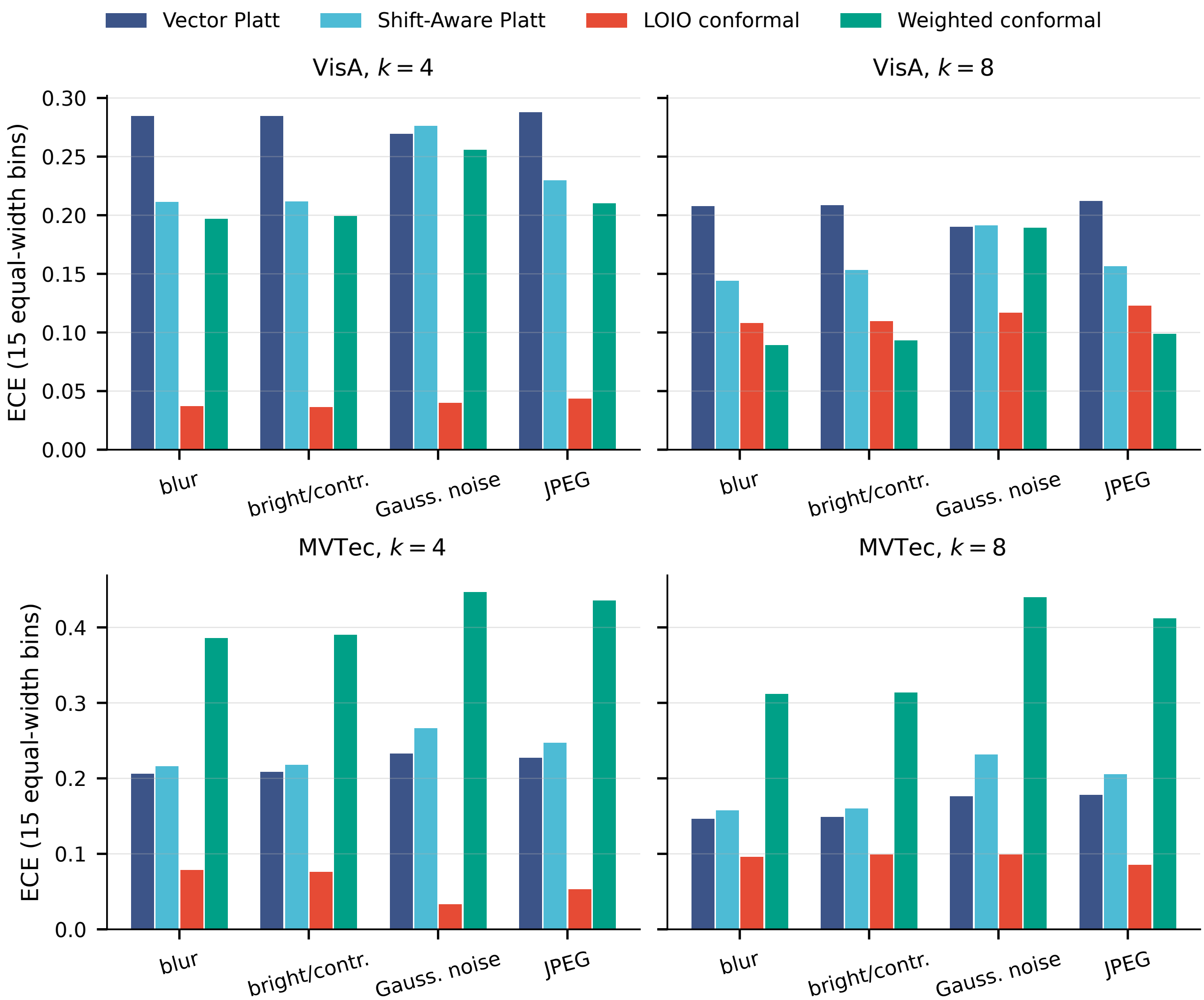

Figure 3: ECE by corruption and reliability route on full VisA (top) and full MVTec (bottom) at k=4 and k=8. Lower values indicate better calibration under the reported ECE definition.

reduces FAR to 0.013/0.040/0.098 but also lowers power to 0.091/0.201/0.415 (Table 12). This transfer direction indicates an empirical trade-off between power and FAR, rather than a general safety property of foreign archives.

5.7. Secondary Deployment and Detector Diagnostics

The risk–coverage analysis in Figure 5 ranks images by predictive entropy. Abstaining from the most uncertain 20% reduces LOIO ECE from 0.068 to 0.030 on full MVTec and from 0.077 to 0.075 on VisA (0.066 at 70% coverage). This serves as a selective-operation diagnostic, not a formal certificate.

Table 13 evaluates protocol-locked representative routing. Fixed LOIO reduces ECE relative to Vector Platt without validation-label expert selection in every tested protocol. This supports using a simple fixed target-only route as the main baseline; learned or weighted routing remains an ablation.

Table 14 separates localization from robustness. The low-storage subspace detector matches or exceeds the controlled memory bank on pixel AUROC at every k and on AUPRO from $k = 4$. By contrast, a label-aware PCA-residual FGSM surrogate reduces image AUROC from 0.94 to 0.16 at $\varepsilon = 2/255$, below chance because the score ordering is inverted. The drop is non-monotonic, recovering to 0.45 at $8/255$, which indicates sensitivity to the surrogate objective itself. These are fragility diagnostics and do not support an adversarial-robustness claim.

6. Limitations

The proposed reliability framework is not a universal accuracy replacement or an MVTec AUROC state-of-the-art claim. Official SubspaceAD remains a strong ranking

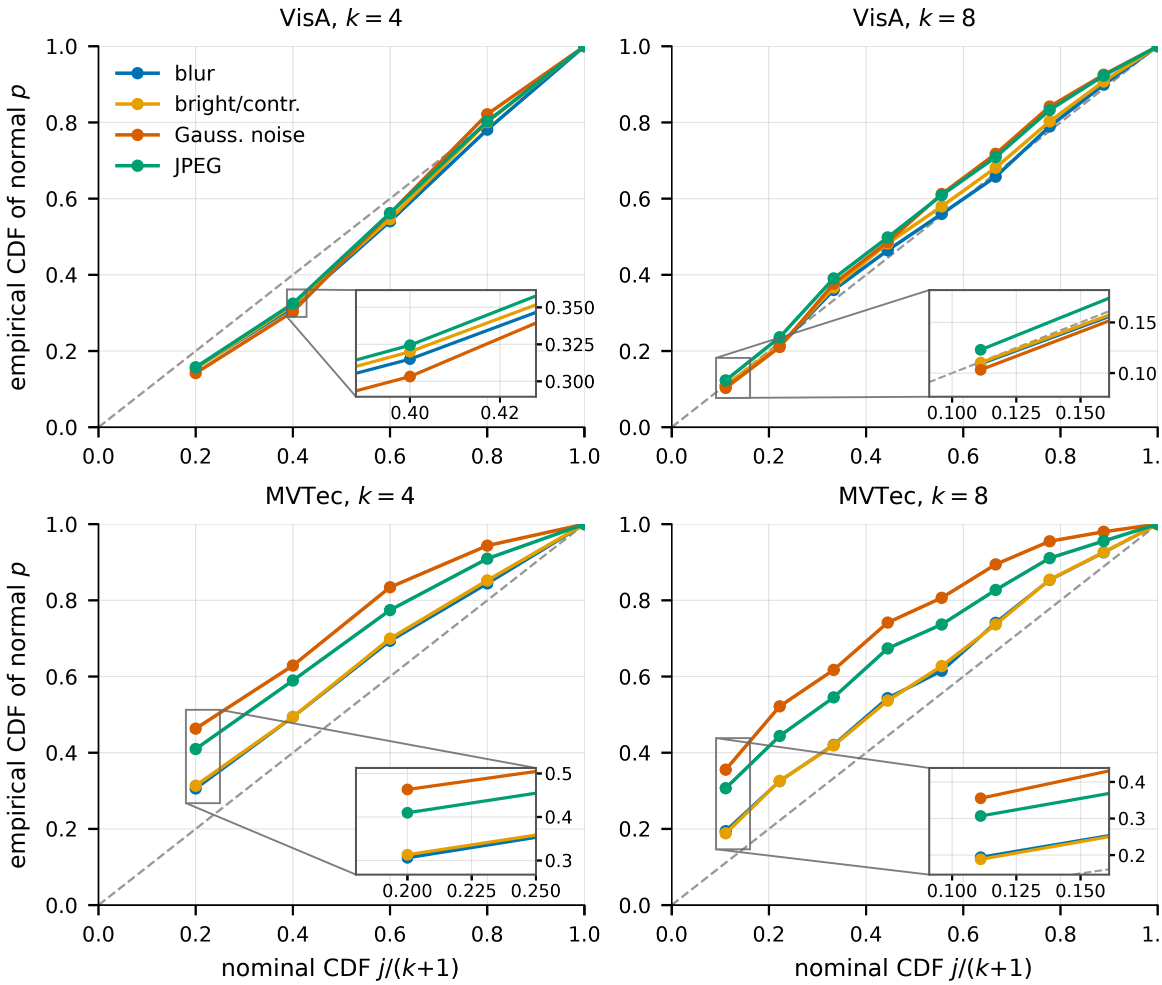

Figure 4: Empirical CDF of normal-image LOIO p-values at each attainable point $j/(k+1)$ versus the idealized exchangeable reference (dashed diagonal), by corruption. Points above the diagonal indicate excess false alarms relative to the reference; lower-right insets enlarge the first informative grid region.

baseline; our contribution concerns compact calibration, operational diagnostics, and source-assisted thresholding.

Neither the framework nor CRESS is adversarially robust. Label-aware PCA-residual FGSM stress tests cause severe degradation and depend on the attack objective, so corruption robustness must not be extrapolated to adversarial robustness.

Conformal validity depends on representative normal support data. LOIO residuals and full-support test residuals are asymmetric, and the normal p-values lie on a finite grid. Section 5.5 therefore reports clustered, discrete-reference diagnostics rather than treating a pooled iid uniformity test as exact. Without the relevant exchangeability conditions, target false-alarm control is not guaranteed.

The historical CRESS tables are empirical evidence only: candidate selection and source assessment were not independent, and their bootstrap intervals were pointwise rather than simultaneous. The frozen nested rerun resolves this ambiguity unfavorably for the strongest claim. Category-level certification selects $\tau = 0$ in every gate cell because three or four certification categories are far below both the Hoeffding feasibility requirement and the multiplicity-free distribution-free all-zero lower bound. The latter assumes a deterministic uniformly valid upper bound based only on iid bounded category losses; additional parametric or hierarchical structure could change the requirement but would need an independently justified model. Fixed-archive image certificates are nonzero in some settings but target a different estimand and cannot be promoted to new-category evidence. Therefore the present work does not establish a deployable certified CRESS alarm. Matched-condition routing additionally assumes condition metadata, and source certification would still require dominance or category exchangeability to transfer to a fixed target.

Finally, ECE for conformal-derived scores is prevalence-sensitive. Calibration numbers on anomaly-balanced benchmarks should not be interpreted as deployment reliability; false-alarm rate, power, precision, and risk–coverage are the primary operational summaries.

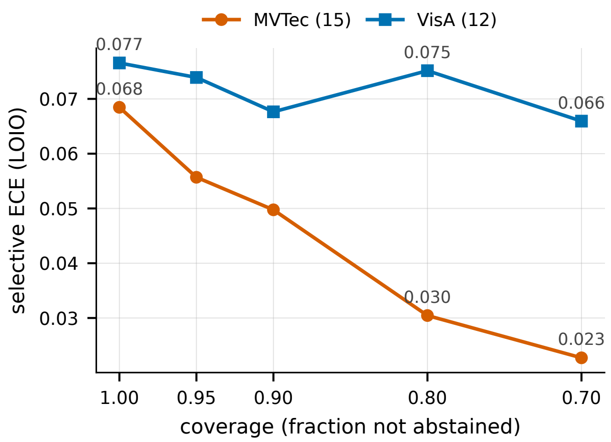

Figure 5: LOIO risk–coverage curves with entropy-ranked abstention on full MVTec and VisA. Coverage is the retained image fraction; risk is ECE on the retained subset.

7. Conclusion and future work

We introduced an auditable separation between low-storage anomaly ranking and few-shot alarm reliability, together with CRESS as the source-assisted cross-category procedure. A target-only alarm built from k calibration scores cannot fire below $\alpha = 1/(k+1)$, and corruption can make its observed false-alarm behavior anti-conservative. Source normals can produce useful sub-floor empirical operating points, as the historical CRESS diagnostics show, but the strict nested experiment establishes a sharper boundary: category-level Hoeffding certification returns the zero threshold in all 960 gate cells across MVTec, VisA, and two external-transfer directions. The conclusion is not specific to Hoeffding's looseness: even a multiplicity-free deterministic distribution-free 95% category-risk upper bound needs at least 14 all-zero categories to reach $\alpha = 0.20$, versus the three or four available. Image-level fixed-archive certificates are sometimes nonzero but cannot replace that failed estimand. CRESS should therefore be read as a source-assisted reliability-estimation and threshold-certification procedure, not as an AUROC state-of-the-art, adversarial-robustness, or certified target-control claim.

Future work should prioritize category-rich source archives or a pre-specified tighter category-risk construction with demonstrable finite-sample validity; adding seeds or source images cannot solve a shortage of independent categories. A separately evaluated condition detector and an audited vision–language baseline remain secondary extensions.

CRediT authorship contribution statement

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed during the current study are publicly available: MVTec AD (`https://www.mvtec.com/company/research/datasets/mvtec-ad`) and VisA (`https://github.com/amazon-science/spot-diff`). The final review package will include derived per-image prediction tables, evaluation summaries, sampling and source-partition manifests, and checksums; the reviewer-accessible repository identifier must be inserted before submission.

Code availability

The experimental code includes the conformal protocols, calibrators, corruption pipeline, artifact audits, and table/figure generators. A reviewer-accessible repository and immutable commit identifier must be inserted here before submission.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used OpenAI Codex for code review, test generation, statistical-claim auditing, and language editing. No experimental observation was generated or altered by the tool. The authors reviewed the resulting code and text and take responsibility for the content.

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A. Supplementary tables

This appendix collects supporting tables summarized in the main text: the per-corruption VisA ECE breakdown and standard-calibrator comparison (Tables 7 and 8); the historical, non-independent CRESS sensitivity studies (Tables 9–12); protocol-routing, localization, and fragility diagnostics (Tables 13 and 14); the aggregation-sensitivity ablation (Table 15); and the novelty-positioning summary (Table 16).

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Table 7
ECE by reliability route on the full VisA corruption benchmark, pooled over classes and seeds and reported separately for each k–corruption cell. Lower is better, and bold marks the lowest ECE in each cell. Figure 3 gives the corresponding visual comparison for VisA and MVTec.

k	Corruption	Vector	Shift-aware	LOIO	Weighted
4	blur	0.284	0.211	**0.037**	0.197
	brightness/contrast	0.285	0.212	**0.036**	0.199
	gaussian noise	0.270	0.276	**0.040**	0.256
	jpeg	0.288	0.230	**0.043**	0.210
8	blur	0.208	0.144	0.108	**0.089**
	brightness/contrast	0.209	0.153	0.109	**0.093**
	gaussian noise	0.190	0.191	**0.117**	0.189
	jpeg	0.212	0.156	0.123	**0.099**

Table 8
LOIO conformal reliability versus label-free calibrators on the historical full corruption benchmarks. Entries are per-cell ECE mean±standard deviation; Δ is LOIO minus baseline (negative favors LOIO), and "LOIO better" is its per-cell win fraction.

dataset / k	baseline	base ECE	LOIO ECE	Δ	LOIO better
MVTec k=4	Vector Platt	0.223±0.095	**0.120±0.054**	-0.104	84%
	Shift-Aware Platt	0.241±0.104	**0.120±0.054**	-0.121	88%
	Temperature scaling	0.240±0.104	**0.120±0.054**	-0.120	87%
	Isotonic regression	0.222±0.088	**0.120±0.054**	-0.103	87%
	Histogram binning	0.229±0.099	**0.120±0.054**	-0.110	84%
	Scalar Platt	0.268±0.127	**0.120±0.054**	-0.148	93%
MVTec k=8	Vector Platt	0.167±0.065	**0.135±0.055**	-0.033	66%
	Shift-Aware Platt	0.192±0.088	**0.135±0.055**	-0.057	75%
	Temperature scaling	0.177±0.080	**0.135±0.055**	-0.042	69%
	Isotonic regression	0.157±0.069	**0.135±0.055**	-0.023	55%
	Histogram binning	0.162±0.080	**0.135±0.055**	-0.027	61%
	Scalar Platt	0.256±0.132	**0.135±0.055**	-0.121	91%
VisA k=4	Vector Platt	0.286±0.060	**0.131±0.063**	-0.155	97%
	Shift-Aware Platt	0.237±0.087	**0.131±0.063**	-0.106	90%
	Temperature scaling	0.312±0.085	**0.132±0.059**	-0.180	98%
	Isotonic regression	0.239±0.073	**0.132±0.059**	-0.106	90%
	Histogram binning	0.301±0.087	**0.132±0.059**	-0.168	96%
	Scalar Platt	0.304±0.145	**0.132±0.059**	-0.171	90%
VisA k=8	Vector Platt	0.199±0.063	**0.151±0.054**	-0.048	81%
	Shift-Aware Platt	0.162±0.075	**0.151±0.054**	-0.011	51%
	Temperature scaling	0.247±0.089	**0.158±0.051**	-0.089	75%
	Isotonic regression	0.194±0.068	**0.158±0.051**	-0.035	62%
	Histogram binning	0.242±0.097	**0.158±0.051**	-0.083	69%
	Scalar Platt	0.418±0.100	**0.158±0.051**	-0.259	98%

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Table 9
Historical matched-condition CRESS versus the target-only LOIO anchor on 15 label-stratified MVTec classes (k=4, seeds 0–2). Threshold selection and source assessment are not independent. Bold marks cells with nonzero power and FAR $\leq \alpha$+0.02.

corruption	α	CRESS (historical) FAR	CRESS (historical) power	CRESS (historical) prec.	Target-only anchor FAR	Target-only anchor power	Target-only anchor prec.
clean	0.05	**0.050**	**0.267**	0.928	0.000	0.000	–
	0.10	**0.100**	**0.470**	0.957	0.000	0.000	–
	0.20	**0.209**	**0.768**	0.914	0.267	0.863	0.898
blur	0.05	**0.048**	**0.242**	0.915	0.000	0.000	–
	0.10	**0.099**	**0.468**	0.950	0.000	0.000	–
	0.20	**0.207**	**0.760**	0.904	0.269	0.860	0.895
bright/contr.	0.05	**0.050**	**0.254**	0.914	0.000	0.000	–
	0.10	**0.101**	**0.467**	0.954	0.000	0.000	–
	0.20	**0.205**	**0.769**	0.905	0.268	0.863	0.898
Gauss. noise	0.05	**0.048**	**0.184**	0.934	0.000	0.000	–
	0.10	**0.105**	**0.295**	0.929	0.000	0.000	–
	0.20	0.226	0.495	0.914	0.460	0.860	0.844
JPEG	0.05	**0.051**	**0.134**	0.911	0.000	0.000	–
	0.10	0.121	0.381	0.936	0.000	0.000	–
	0.20	0.235	0.633	0.917	0.434	0.867	0.854

Table 10
Historical matched-condition CRESS versus the target-only LOIO anchor on 15 label-stratified MVTec classes (k=8, seeds 0–2). As in the k=4 historical analysis, threshold selection and source assessment are not independent. Bold marks cells with nonzero power and FAR $\leq \alpha$+0.02.

corruption	α	CRESS (historical) FAR	CRESS (historical) power	CRESS (historical) prec.	Target-only anchor FAR	Target-only anchor power	Target-only anchor prec.
clean	0.05	**0.053**	**0.764**	0.976	0.000	0.000	–
	0.10	**0.110**	**0.833**	0.936	0.000	0.000	–
	0.20	**0.216**	**0.898**	0.917	**0.182**	**0.903**	0.928
blur	0.05	**0.050**	**0.751**	0.976	0.000	0.000	–
	0.10	**0.110**	**0.813**	0.957	0.000	0.000	–
	0.20	0.221	0.900	0.917	**0.199**	**0.904**	0.922
bright/contr.	0.05	**0.051**	**0.770**	0.976	0.000	0.000	–
	0.10	**0.107**	**0.831**	0.958	0.000	0.000	–
	0.20	0.223	0.895	0.916	**0.185**	**0.904**	0.926
Gauss. noise	0.05	**0.035**	**0.155**	0.962	0.000	0.000	–
	0.10	0.138	0.398	0.951	0.000	0.000	–
	0.20	0.232	0.749	0.917	0.392	0.904	0.866
JPEG	0.05	0.071	0.296	0.972	0.000	0.000	–
	0.10	0.141	0.588	0.959	0.000	0.000	–
	0.20	0.260	0.826	0.918	0.380	0.911	0.875

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Table 11
Historical comparison of smoothed conformal p-values and non-independent CRESS below the target-only floor (k=4, all classes, seeds 0–2, five pooled conditions). Randomized rates are exact expectations over the auxiliary randomization. Bold marks FAR $\leq \alpha+0.02$.

		Randomized p-value		CRESS (historical)	
dataset	α	FAR	power	FAR	power
MVTec	0.05	0.085	0.216	**0.049**	0.216
	0.10	0.170	0.431	**0.105**	0.416
VisA	0.05	**0.036**	0.150	**0.051**	0.144
	0.10	**0.072**	0.300	**0.095**	0.351

Table 12
Historical matched-condition CRESS on 12 label-stratified VisA classes (k=4, seeds 0–2, five pooled conditions), using either within-VisA or MVTec source categories. Threshold selection and source assessment are not independent in this historical analysis. Bold marks cells with nonzero power and FAR $\leq \alpha+0.02$.

source pool	α	FAR	power	prec.	no-harm
VisA (within)	0.05	**0.051**	**0.144**	0.847	89%
	0.10	**0.095**	**0.351**	0.843	84%
	0.20	**0.192**	**0.637**	0.810	78%
MVTec (cross-dataset)	0.05	**0.013**	**0.091**	0.936	97%
	0.10	**0.040**	**0.201**	0.881	93%
	0.20	**0.098**	**0.415**	0.852	93%

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Table 13
Protocol-locked representative comparison of fixed LOIO conformal and Vector Platt across cross-dataset, leave-one-category-out (LOCO), and within-split settings. ΔECE is LOIO minus Vector ECE, so negative values favor LOIO; no-harm is the fraction of cells in which LOIO does not increase ECE.

Protocol	Vector ECE	LOIO ECE	Δ ECE	No-harm
MVTec→VisA	0.301	**0.099**	−0.203	100%
VisA→MVTec	0.304	**0.102**	−0.202	100%
LOCO	0.304	**0.116**	−0.188	100%
Within split	0.361	**0.154**	−0.207	100%

Table 14
MVTec pixel-level localization and FGSM fragility diagnostics, averaged over classes and five seeds. The upper block reports pixel AUROC and AU-PRO; the lower block reports clean and perturbed image AUROC under the label-aware PCA-residual surrogate.

Pixel localization			
k	Method	pixel AUROC	AU-PRO
1	PCA64 (ours)	**0.944**	0.820
	Controlled DINOv2 NN	0.940	**0.845**
4	PCA64 (ours)	**0.957**	0.862
	Controlled DINOv2 NN	0.951	0.862
8	PCA64 (ours)	**0.960**	**0.876**
	Controlled DINOv2 NN	0.952	0.863
FGSM fragility diagnostic (image AUROC)			
ε	k	clean → FGSM	rel. drop
2/255	4	0.937 → 0.158	83%
	8	0.945 → 0.175	82%
8/255	4	0.937 → 0.446	52%
	8	0.945 → 0.446	53%

Table 15
Aggregation-sensitivity ablation: clean image AUROC averaged over representative classes and seeds 0–2 for max and top-ρ mean pooling. Bold marks the best aggregator in each row.

dataset	k	max	top-0.5%	top-1%	top-2%	top-5%
MVTec	4	0.9571	0.9638	0.9667	**0.9693**	0.9621
	8	0.9578	0.9685	0.9740	**0.9759**	0.9727
VisA	4	0.9039	**0.9101**	0.9038	0.8944	0.8806
	8	0.9101	**0.9200**	0.9147	0.9071	0.8941

Table 16
Novelty positioning: each ingredient has established prior art, and this paper's contribution begins after it. The right column marks where prior work ends and our claims start.

Work family	Established prior art	Our contribution starts after it
AnomalyDINO	DINOv2 few-shot patch/memory-bank AD	Reliability of the alarms such detectors raise
SubspaceAD	DINOv2 PCA/subspace residual ranking	Operational reliability audit downstream of the fixed residual
Calibration/robustness studies	ECE, Platt scaling, FGSM fragility	Conformal-native audit; full calibrator toolbox under one label-free protocol
Conformal AD	P-values from nonconformity scores	The few-shot floor, empirical source routing, and finite-category certificate feasibility
SAGE-style routing	Gated shared/specialized experts	Fixed no-label reliability routing and its failure boundaries